



London Ambulance Service



NHS Trust

Anatomy and Physiology – Part Component 2

Musculoskeletal System



Musculoskeletal System - Part Component 2

Objective

- Explain structure and function of the musculoskeletal system
- Have an understanding of how joints, bones, muscles, ligaments and tendons work together



Musculoskeletal System - Part Component 2

Overview

- Bones
- Muscle tissue
- Joints and connective tissue



Musculoskeletal System - Part Component 2

Functions of the Skeletal system

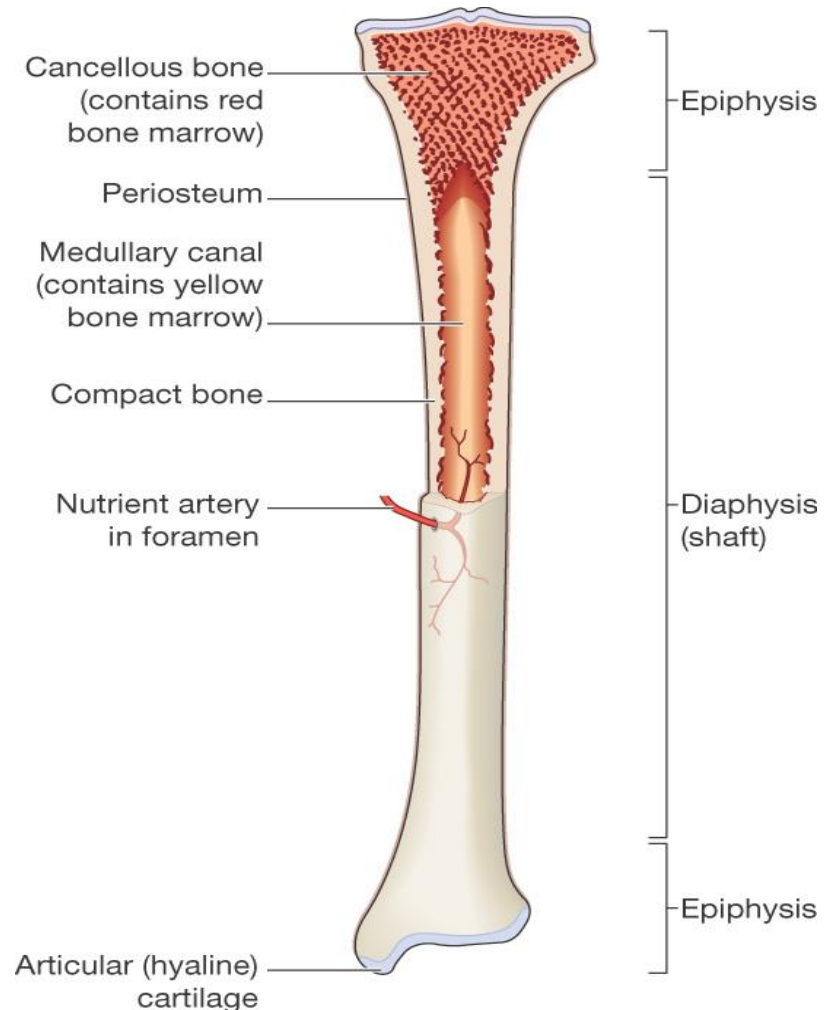
1. Support
2. Protection
3. Movement
4. Mineral storage
5. Blood cell production
6. Energy storage



Bone type	Description	Example
Long	Consist of a shaft and two extremities. They are longer than they are wide.	Humerus
Short	No shafts or extremities, and are diverse in shape and size	Carpals, (wrist),
Flat	No shafts or extremities, and are diverse in shape and size	Sternum, ribs, most skull bones
Sesamoid	No shafts or extremities, and are diverse in shape and size	Patella, (knee cap)
Irregular	No shafts or extremities, and are diverse in shape and size	Vertebrae, some skull bones

Musculoskeletal System - Part Component 2

Bone Anatomy



© 2006 Elsevier Ltd.

Musculoskeletal System - Part Component 2

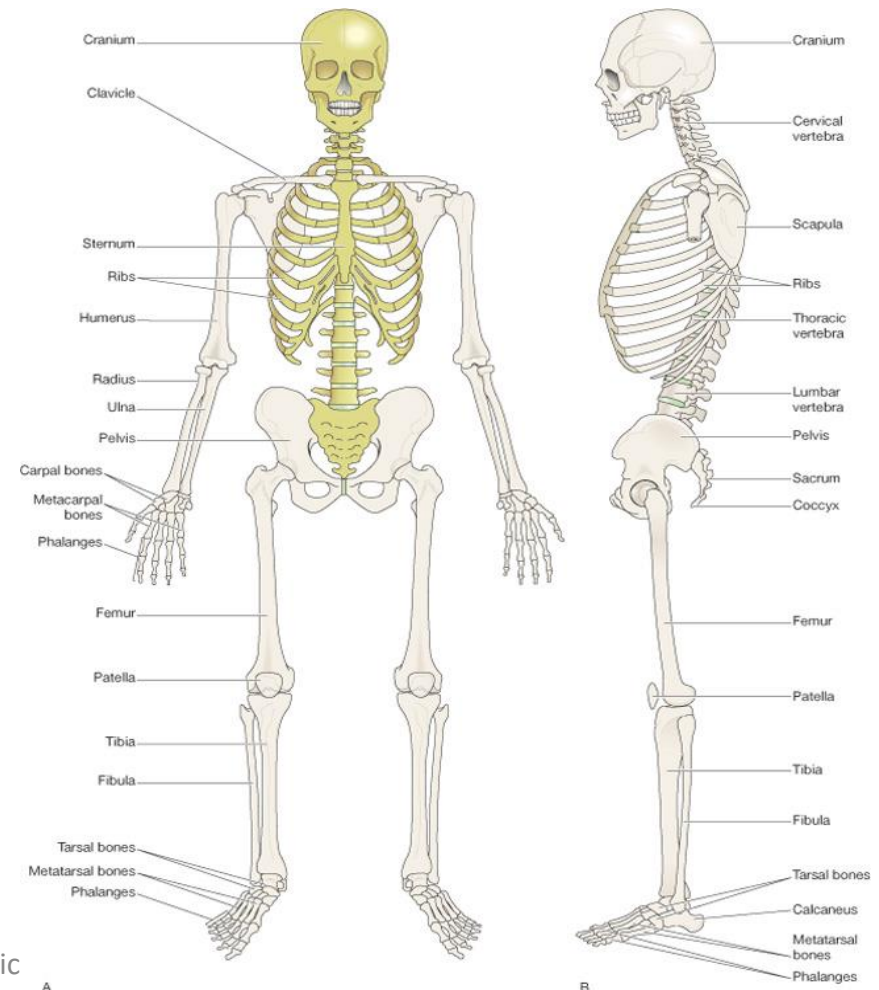
Axial and Appendicular Skeleton

Axial

- 80 bones
- Skull, vertebrae, sternum, ribs

Appendicular

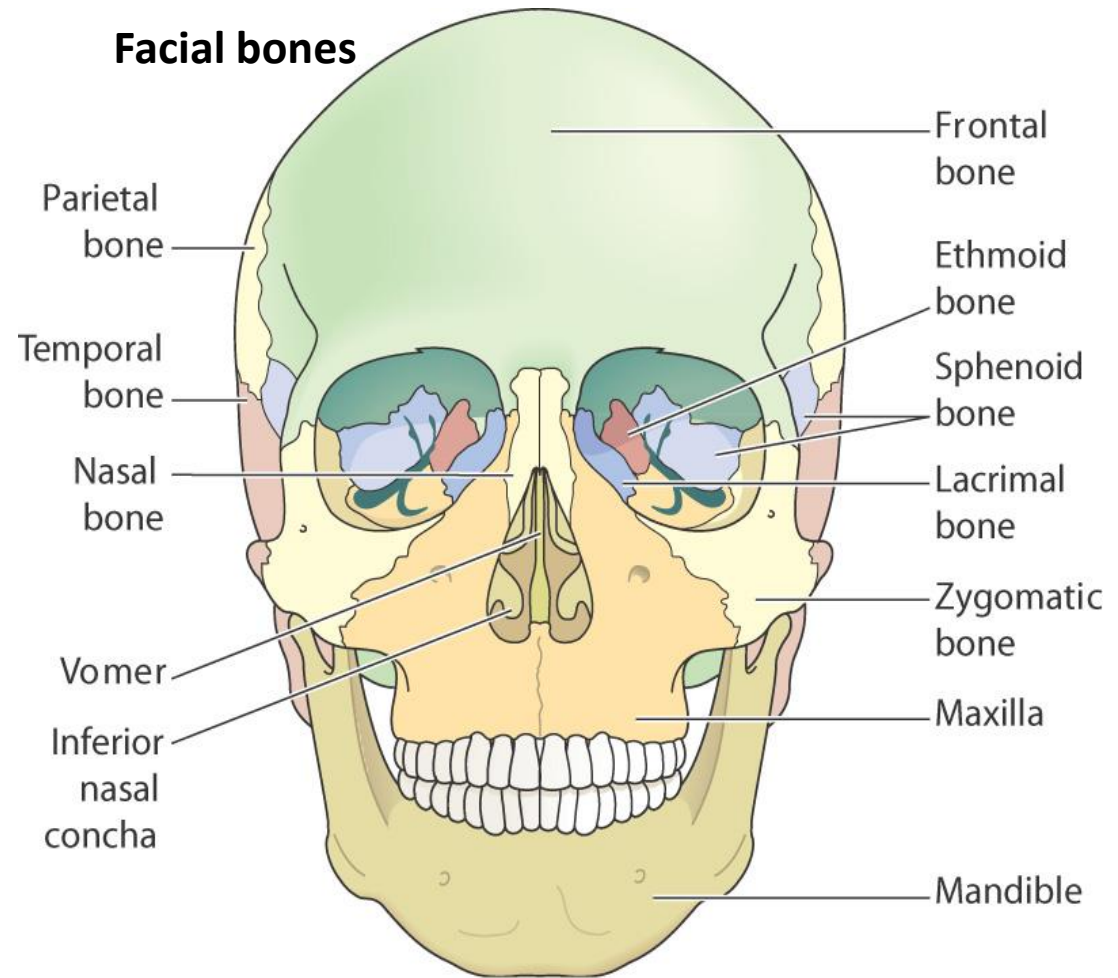
- 126 bones
- Pelvic, shoulder girdles, arm, leg, feet, hand



Musculoskeletal System - Part Component 2

Skull

- Frontal
- Parietal
- Temporal
- Zygoma
- Maxilla
- Mandible

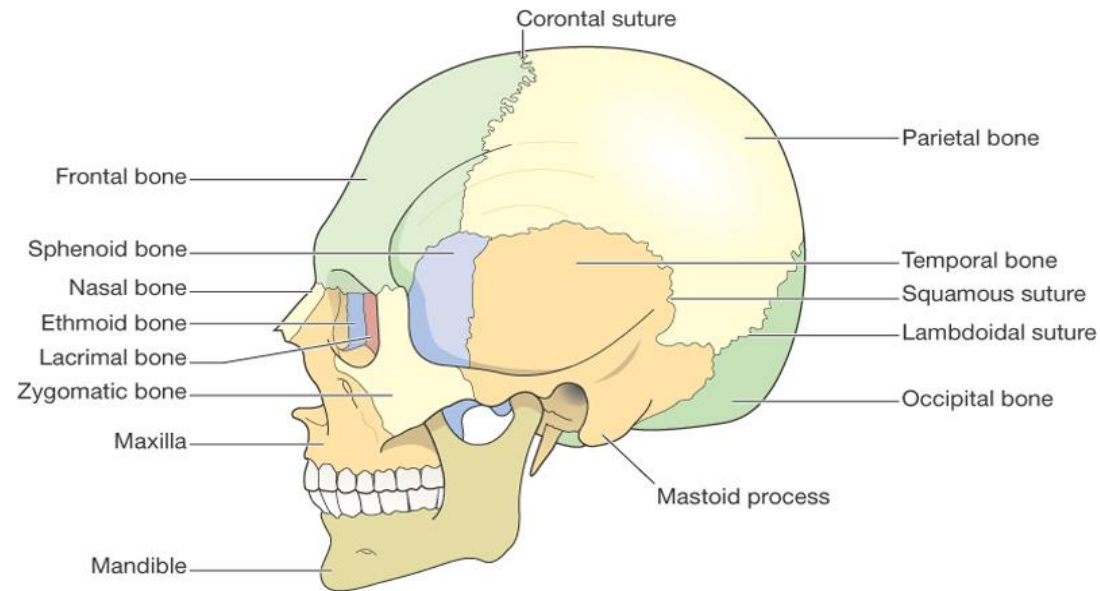


© 2006 Elsevier Ltd.

Musculoskeletal System - Part Component 2

Skull

- Frontal
- Parietal
- Occipital



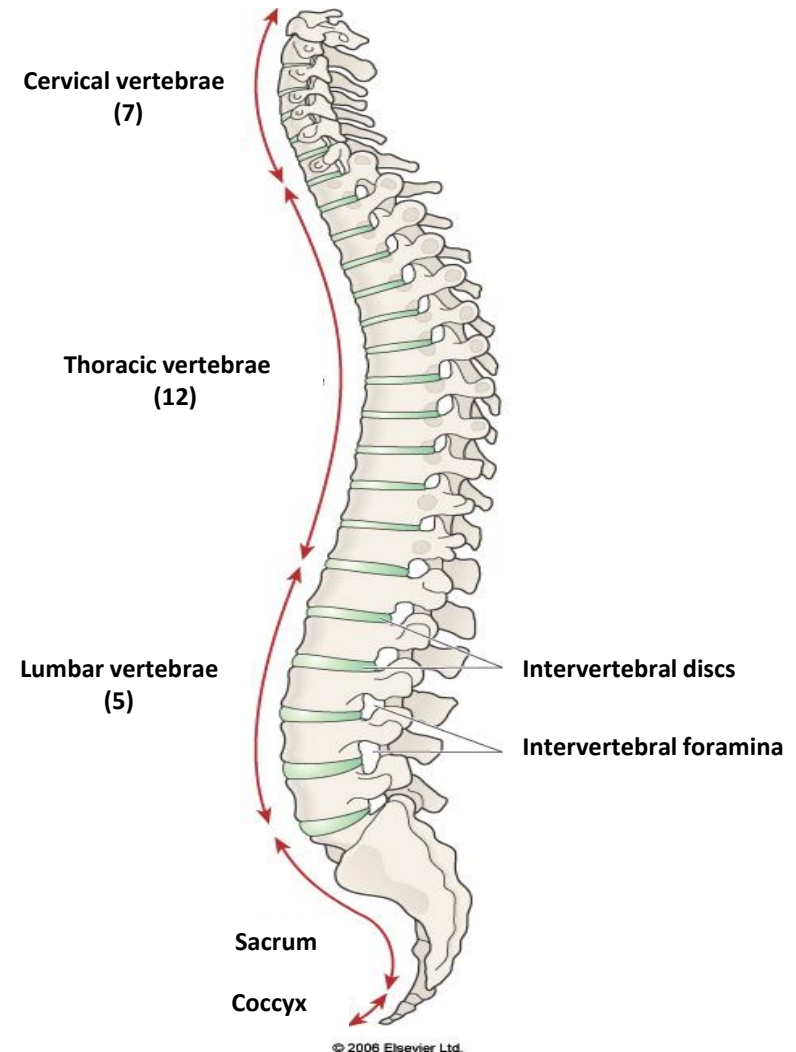
© 2006 Elsevier Ltd.

Musculoskeletal System - Part Component 2

Spine

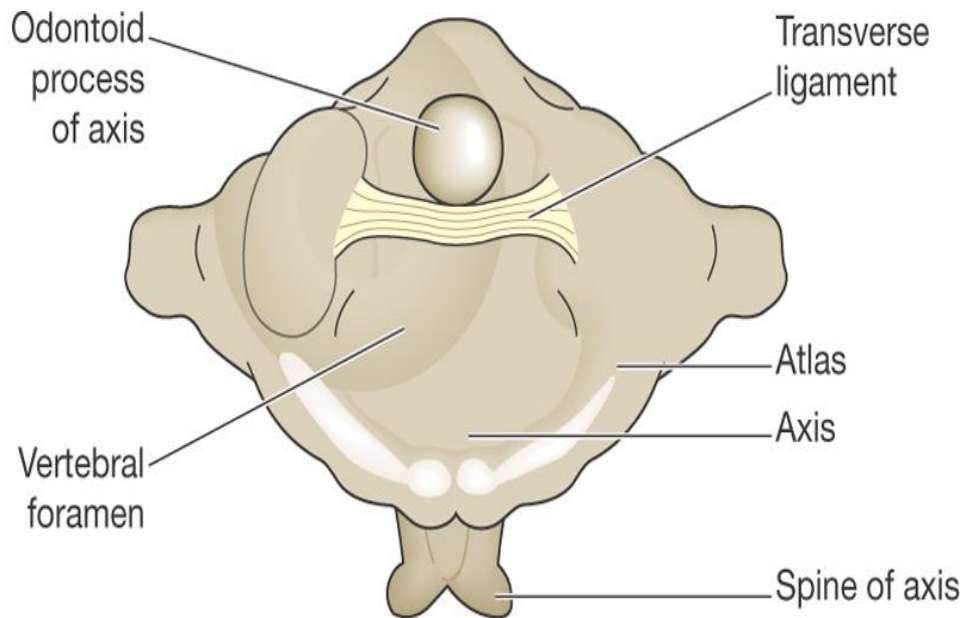
Vertebral column

- Cervical (7)
- Thoracic (12)
- Lumbar (5)
- Sacral (5)
- Coccygeal (4)



Musculoskeletal System - Part Component 2

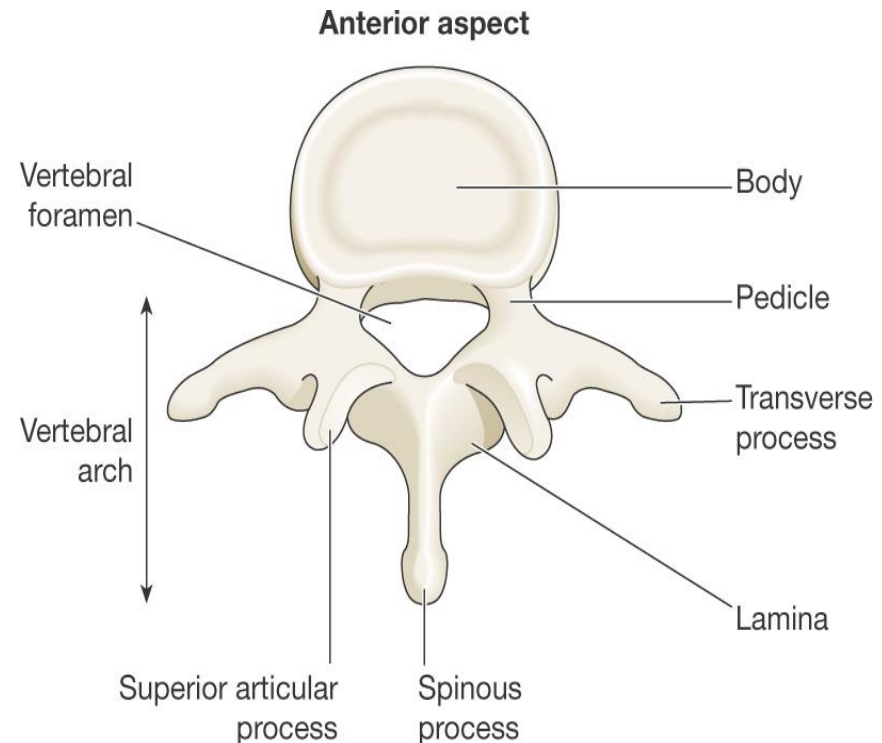
Odontoid process



© 2006 Elsevier Ltd.

Ross and Wilson, 2014, p403

Lumbar Vertebra

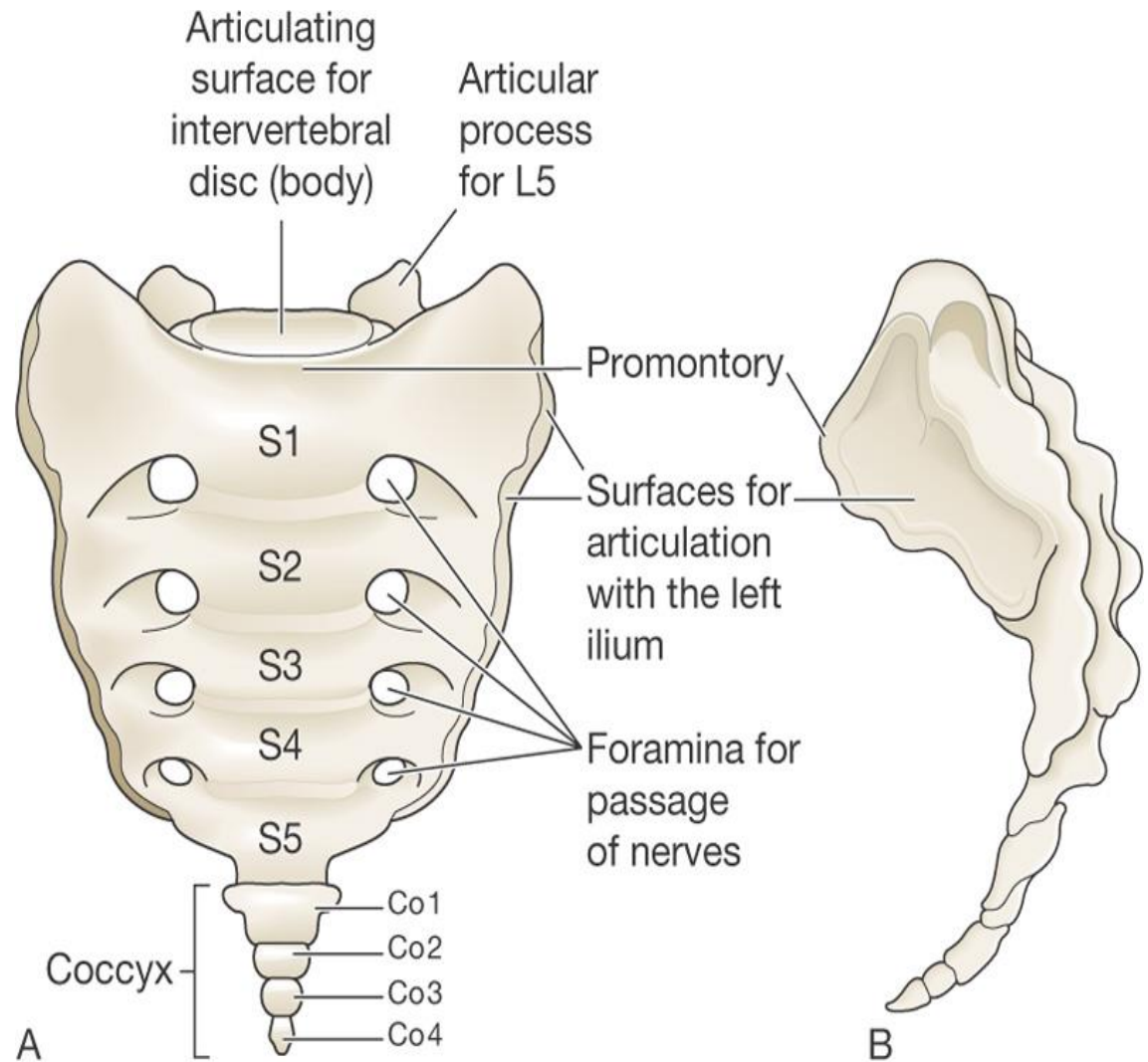


© 2006 Elsevier Ltd.

Ross and Wilson, 2014, p402

Musculoskeletal System - Part Component 2

Sacrum and Coccyx

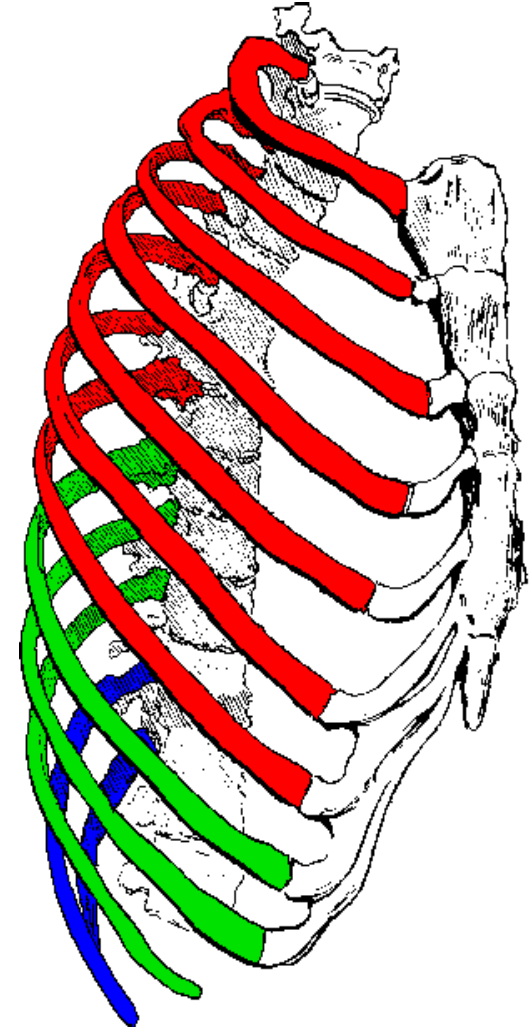


Ross and Wilson, 2014, p404

Musculoskeletal System - Part Component 2

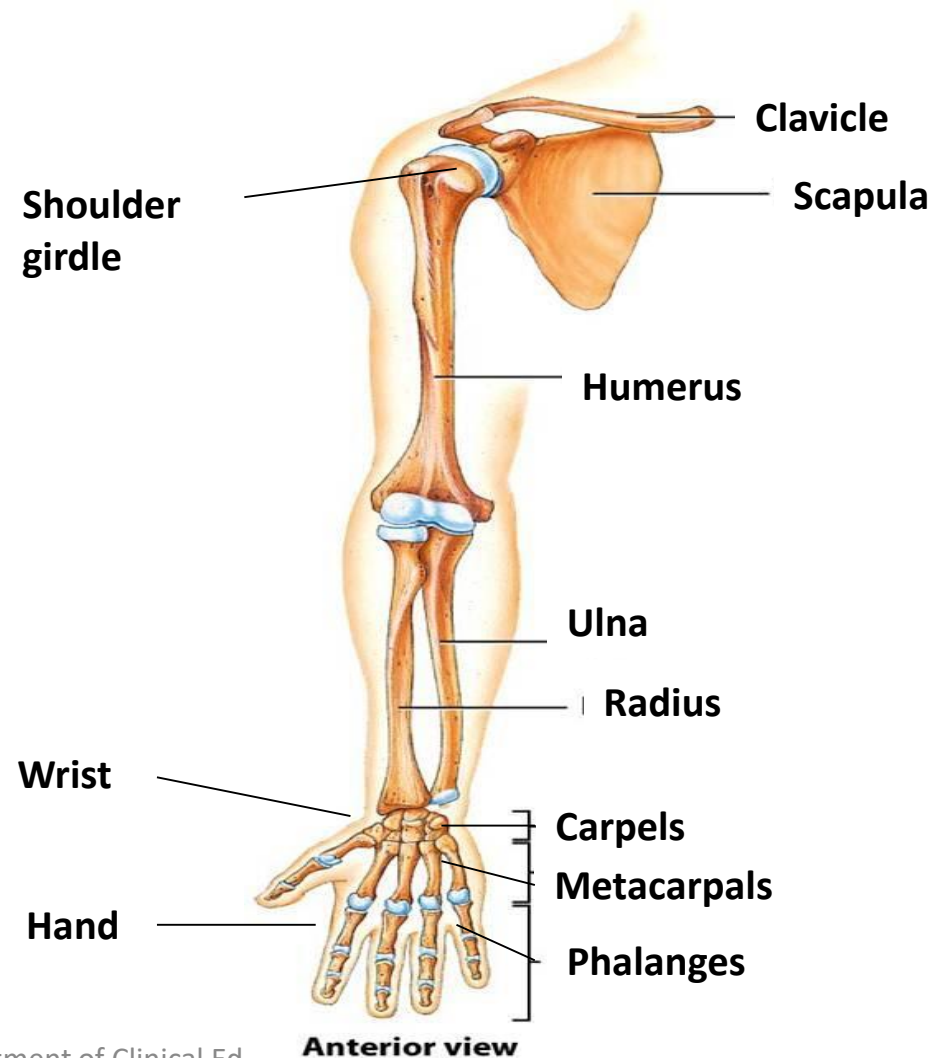
Thorax

- Thoracic Cage
- Thoracic Vertebrae
- Ribs 12 pairs
 - True 7 pairs
 - False 3 pairs
 - Floating 2 pairs
- Sternum



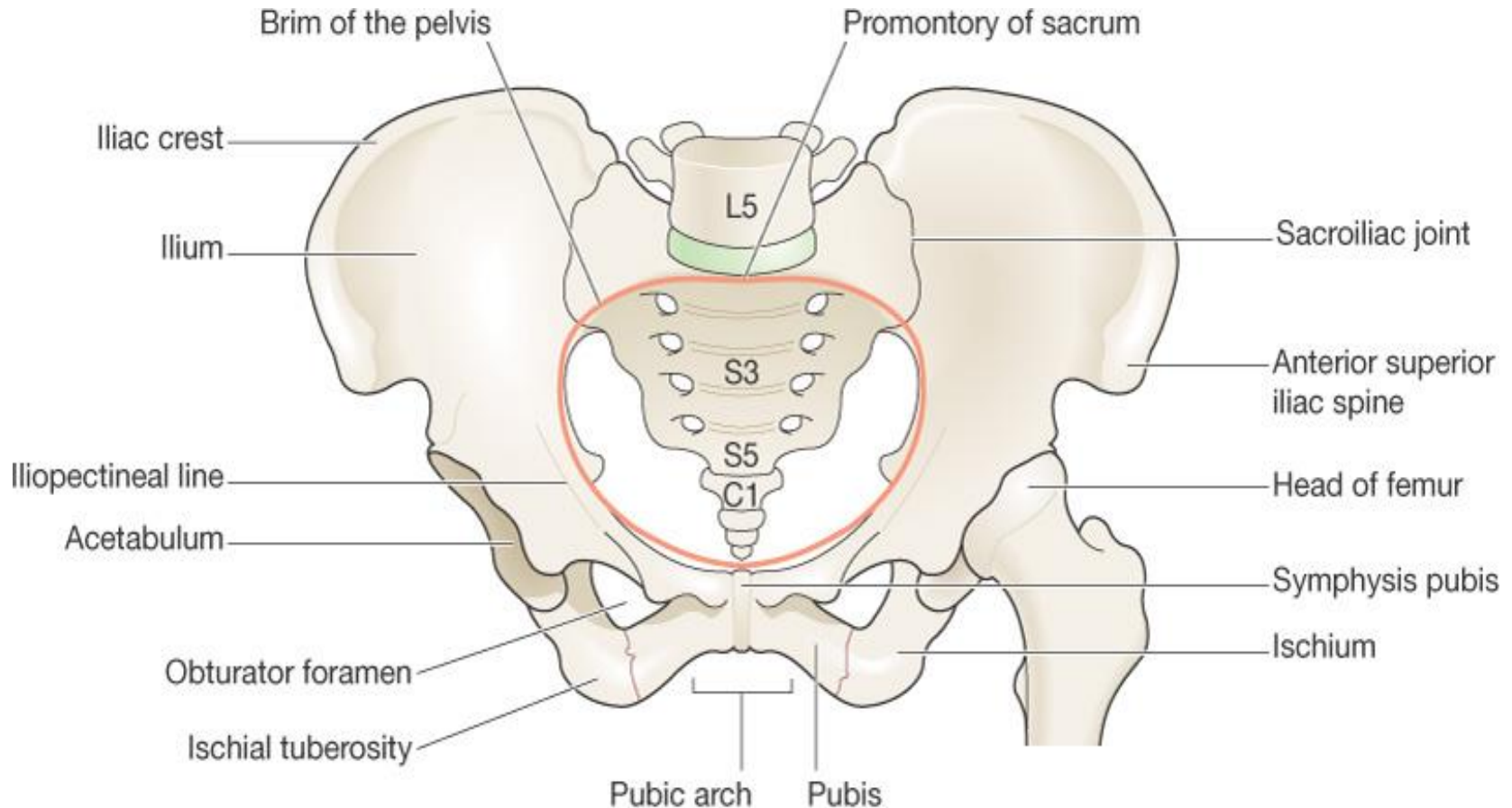
Musculoskeletal System - Part Component 2

Arm , Hand and Shoulder girdle



Musculoskeletal System - Part Component 2

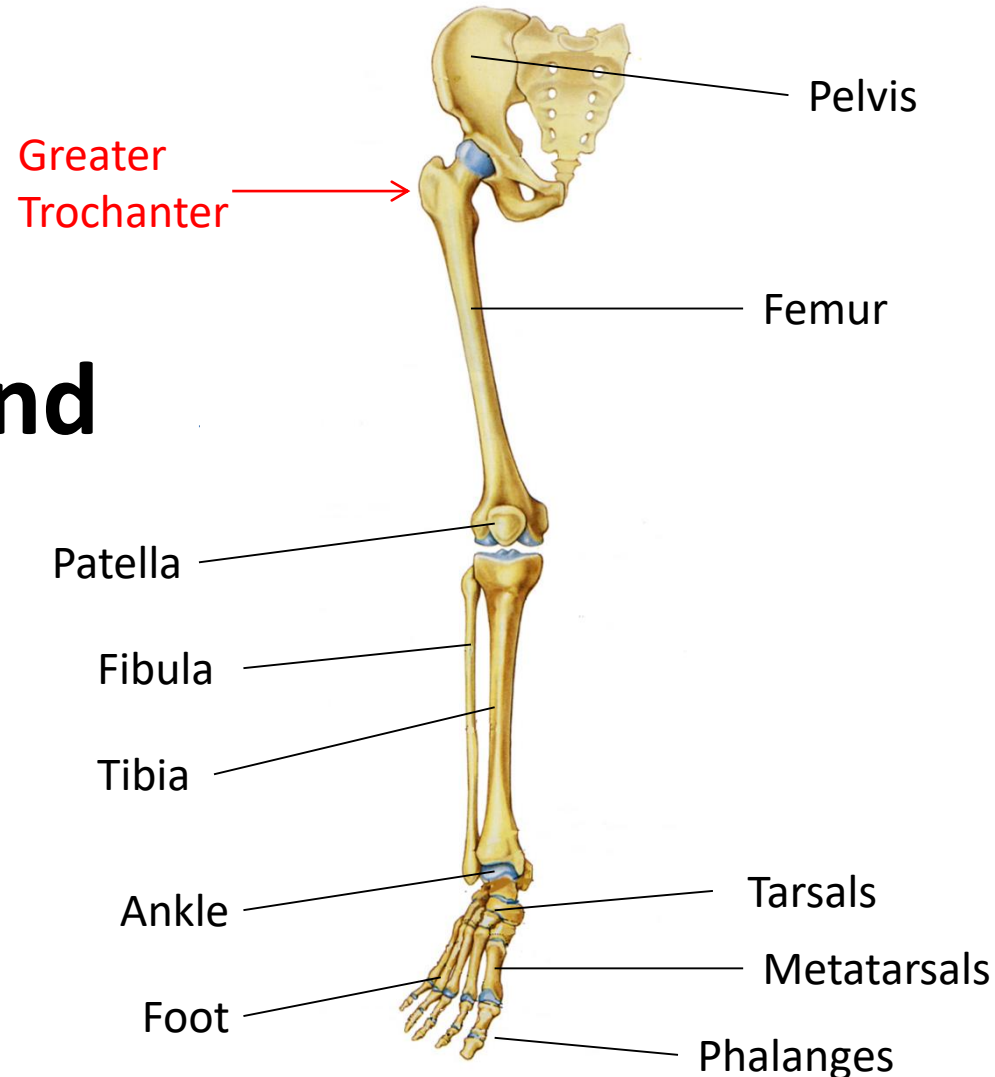
Pelvis



© 2006 Elsevier Ltd.

Musculoskeletal System - Part Component 2

Pelvis, leg and foot





Musculoskeletal System - Part Component 2

Types of muscle

- Skeletal – moves the skeleton, usually voluntary control
- Smooth – not under conscious control
- Cardiac – only found in heart wall



Musculoskeletal System - Part Component 2

Tendons and Ligaments

- Tendon – fibrous tissue which connects muscle to bone
- Ligament – connective tissue which connects bone to bone



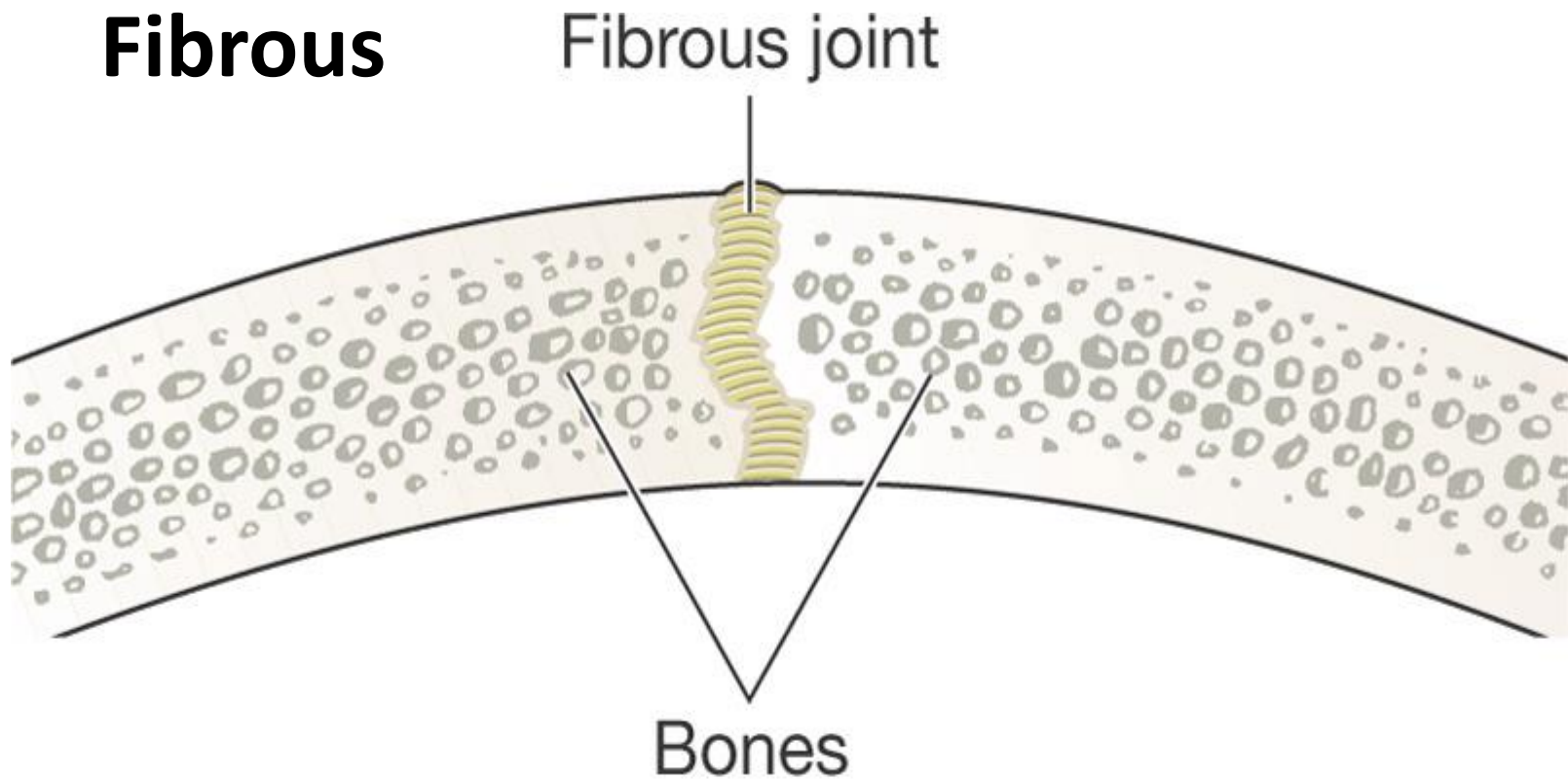
Musculoskeletal System - Part Component 2

Joints

“Joints allow flexibility and movement of the skeleton and allow attachment between bones”.

Ross and Wilson, 2014, p412

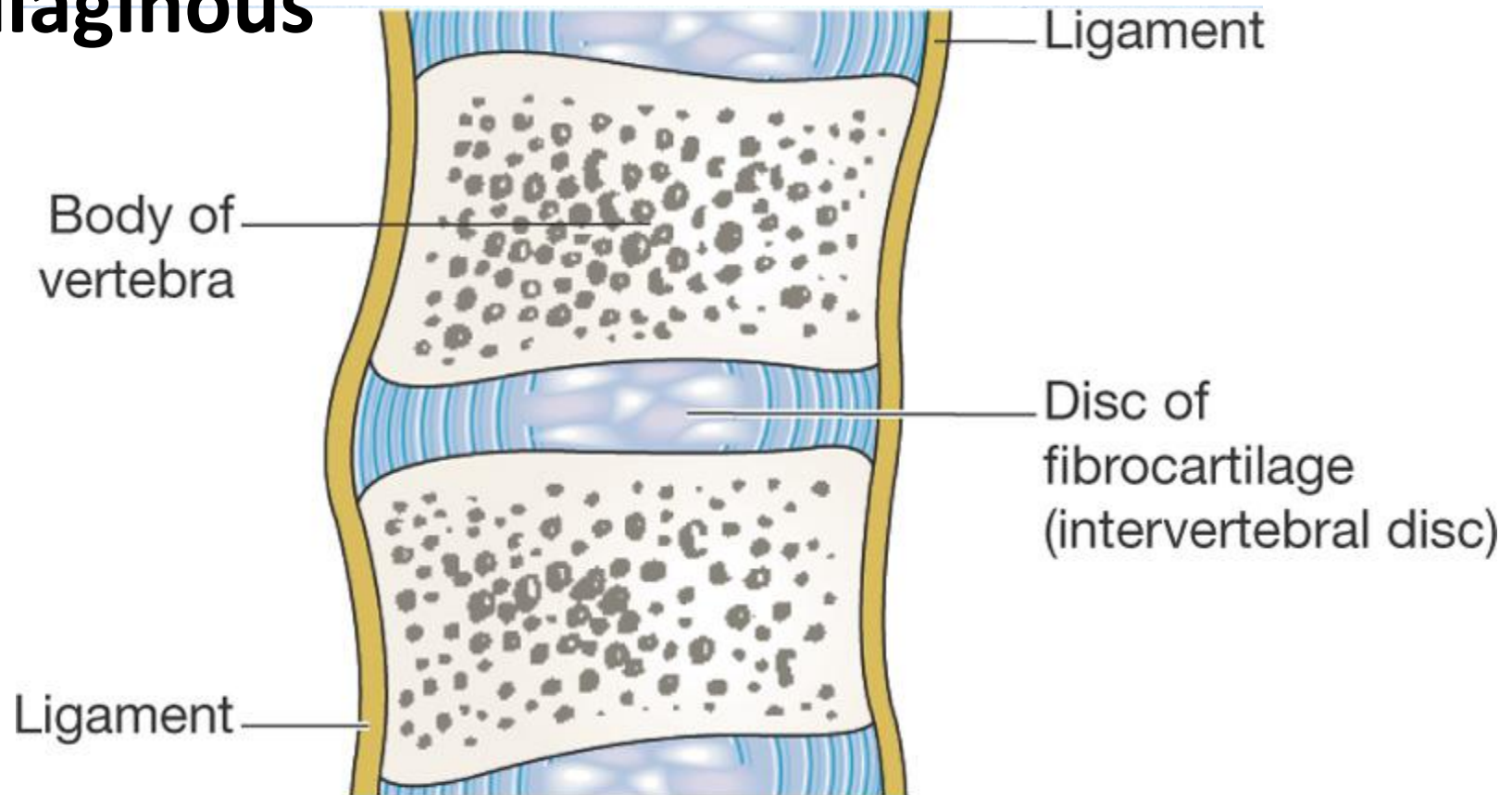
Musculoskeletal System - Part Component 2



© 2006 Elsevier Ltd.

Musculoskeletal System - Part Component 2

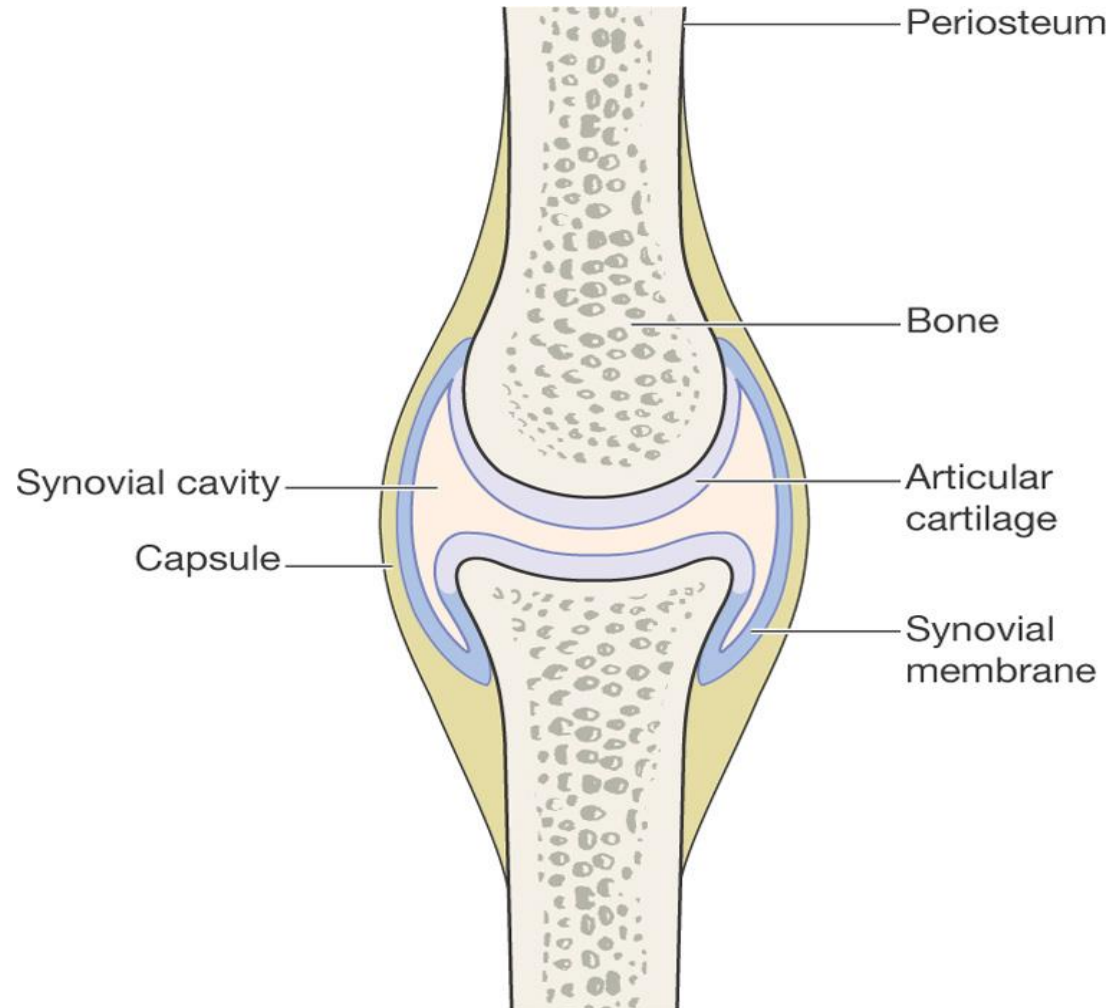
Cartilaginous



© 2006 Elsevier Ltd.

Musculoskeletal System - Part Component 2

Synovial joint



© 2006 Elsevier Ltd.



Musculoskeletal System - Part Component 2

Types of synovial joints

Joint Name	Movement	Example location
Ball and socket	Ball shaped end of bone into cup shaped socket. Wide range of movement	Shoulder and hip
Hinge	Bone ends fit together like a hinge on a door	Elbow and knee
Gliding	Flat or very slightly curved bone surfaces which glide over one another	Between carpal bones in the wrist and tarsal bones in feet



Musculoskeletal System - Part Component 2

Joint Name	Movement	Example location
Pivot	Allow bone to rotate. One bone fits into a hoop shaped ligament which holds it close to another bone	Head rotation
Condylloid	Smooth rounded projection which sits in cup shaped depression in another joint. Allows flexion, extension, abduction, adduction and circumduction	Between metatarsals and phalangeal bones of foot and metacarpals and phalangeal bones of hand.
Saddle	Fit together like person sitting on saddle. Movement like condylloid but with further flexibility	Base of the thumb

Musculoskeletal System - Part Component 2

**Any
Questions ?**

